

Mock Examination, 2023
Session 2022-23

Max. Marks: 50

Time Allowed: 1 hours

Question Number 1

a Name **three** types of bus used in the von Neumann architecture.[3]

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.....

.....

b The contents at a number of addresses are shown below.[5]

Address	Contents
1000 0000	0111 1100
1000 0001	1000 0011
1000 0010	1111 1101
1000 0011	0111 1110
1000 0100	1100 1101
▼ 1111 1100	▼
1111 1101	0110 0011
1111 1110	1010 0111
1111 1111	1111 0011

i Show the contents of the MAR and MDR if we READ the contents of memory location 1 1 1 1 1 1 1 0.

MAR:								
-------------	--	--	--	--	--	--	--	--

MDR:								
-------------	--	--	--	--	--	--	--	--

ii Show the contents of the MAR and MDR if we wish to write 1 1 0 0 1 1 0 1 into memory location 1 0 0 0 0 1 0 0.

MAR:								
-------------	--	--	--	--	--	--	--	--

MDR:								
-------------	--	--	--	--	--	--	--	--

iii If MAR contains 1 1 1 1 1 1 0 0 and MDR contains 0 0 1 1 1 1 0 0, complete the memory contents diagram above.

Question Number 2

a Explain the meaning of the following terms, with reference to the CPU.[2]

i (system) clock cycle:

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.....

ii cache:

.....

.....

b Discuss how changes to the clock speed and to the number of cores can affect the performance of the CPU. Include any advantages and disadvantages, due to these changes, in your discussion.[4]

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Question Number 3

a i Define what is meant by an embedded system.[1]

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ii State **four** of the features you would expect to find in any embedded system.[4]

1

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2

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3

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4

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iii Describe **three** applications that use embedded systems.[3]

1

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2

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3

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Question Number 4

A camera uses an embedded system.[3]

a Name **three** of the tasks controlled by the embedded system in the camera.

- 1
- 2
- 3

b Name **two** applications where a digital camera could be used.[2]

- 1
.....
- 2
.....

Question Number 5

Choose the most suitable input device for each of the following applications. A different device must be given in each case.[7]

Application	Most suitable input device
Entering text and numbers into a word processor or spreadsheet	
Selecting an option or icon from an on-screen menu	
Inputting a user's voice into a computer as part of a voice recognition system	
Converting a hard copy document into an electronic form to be stored in a computer	
Reading a QR code using a smartphone or tablet	
Keying in digits from a barcode which did not scan correctly	
Reading data directly from the surroundings, such as taking a temperature	

Question Number 6

a Give **three** advantages of using digital light projectors when compared to LCD projectors.[3]

- 1
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.....
- 2
.....
.....
- 3
.....
.....

b Give **two** disadvantages of using digital light projectors when compared to LCD projectors.[2]

- 1
.....
.....
- 2
.....
.....

Question Number 7

Inkjet printers and laser printers are used in many offices.

a Give **one** advantage and **one** disadvantage of each type of printer when compared to each other.[4]

Inkjet printer:

Advantage:
.....
.....
Disadvantage:
.....

Laser printer:
 Advantage:.....

 Disadvantage:

Question Number 8

[4]

Eight types of sensor are shown on the left and eight potential sensor applications on the right.

Draw lines to connect each sensor to an appropriate application. Only one application can be assigned to each sensor.

Level sensor	Control/monitor the amount of water in the soil in a greenhouse
Moisture sensor	Used by a mobile phone to change between portrait and landscape modes
Light sensor	Monitor the amount of fuel in a car's petrol tank
Infrared (active) sensor	Turn on the windscreen wipers automatically when rain is detected
Pressure sensor	Used in the anti-lock braking systems in an aeroplane under-carriage
Acoustic sensor	Switch vehicle headlights on or off automatically
Magnetic field sensor	Measuring the weight of a lorry or van at a weigh station
Accelerometer	Pick up the noise of breaking glass in a security system

Question Number 9

a Explain the differences between primary memory and secondary storage.[4]

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b In the following table, tick (✓) which description refers to RAM and which refers to ROM.[3]

Description	RAM (✓)	ROM (✓)
Temporary memory device		
Non-volatile memory		
Data stored in this memory cannot be altered		
Permanent memory device		
Stores data and part of the operating system currently in use		
Can be increased in size to improve the operational speed of a computer		

c A model radio-controlled car contains RAM, ROM and also has a USB connection. The operation of the car is controlled by a remote control which communicates using radio waves.

i State what data or information would be stored on the ROM chip.[2]

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.....

.....

ii State what data or information would be stored on the RAM chip.[2]

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.....

.....

iii Explain why a USB port has also been included in the model car.[2]

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Question Number 10 [7]

Backing storage makes use of magnetic, optical or solid-state technology.

Seven descriptions are given in the table below. Indicate with a tick (✓) which statements refer to magnetic, optical or solid-state technology.

Description	Technology used		
	Magnetic (✓)	Optical (✓)	Solid state (✓)
Makes use of floating gate and control gate technology			
Disk surfaces are made up of tracks and sectors; storage relies on certain properties of the iron oxide coating			
Devices using this technology have no moving parts			
Data is stored in 'pits' and 'lands' on a single spiral track running from the centre outwards			
Disks use the properties of laser light to allow read and write operations			
Data is stored by controlling the movement of electrons within NAND chips			
This technology is affected by strong magnetic and radio fields			

Question Number 11**[5]**

There are ten statements in the table below. Indicate, using a tick (✓) whether each statement is true or false.

Statement	True (✓)	False (✓)
Devices using magnetic media have a very slow data transfer rate		
HDD platters can be made from ceramic, glass or aluminium coated in iron oxide		
HDD surfaces are split up into tracks and sectors		
HDDs use a more reliable technology than solid-state devices, such as SSDs		
DVD-Rs are used to read and write data several times		
DVD and Blu-ray disks can both use dual-layer technology		
SSDs suffer from a high degree of latency		
Solid-state devices, known as flash drives, use NAND chip technology		
SSD data access time is approximately 100 times less than the data access time for HDDs		
Memory cards (such as the SD and XD cards used in cameras) are examples of magnetic media		

Question Number 12**[5]**

a State which network terms are being described.

i A circuit board or chip that allows a device to connect to a network:

.....

ii Type of address that uniquely identifies a device connected to a network:

.....

iii Address assigned by an ISP to a device each time it logs onto the internet:

.....

iv Hardware device that allows data packets to be moved between different networks:

.....

v Hardware or software that sits between a computer and an external network; it monitors traffic to and from the computer:

.....

Question Number 13**[4]**

Tick (✓) the appropriate column, in the following table, to indicate whether the named software is an example of system software or application software.

Software	System (✓)	Application (✓)
Photo editing software		
Graphics manipulation software		
Compiler		
Spreadsheet software		
Printer driver		
QR code reader		
Anti-virus software		
Screensaver		

Question Number 14**[6]**

a Give **two** of the general features of system software.

1

- 2
- b Give **two** of the general features of application software.
- 1
- 2
- c Give **three** examples of utility programs.
- 1
- 2

Question Number 15

[8]

- 3 Use the following list of words and phrases to complete the paragraph below.

Each word or phrase can be used once, more than once or not at all.

» anti-virus	» disk defragmenter	» infected	» security system
» background	» file compression	» locked	» smaller
» blocks	» file management	» quarantine	» tracks
» bootstrap	» head movements	» screensavers	» user
» contiguous	» heuristic checking	» sectors	» utility

Running software in the on a computer will constantly check for virus attacks. Many utilities check software for certain behaviour which would indicate a possible virus; this is known as Any possible files or programs which are infected are put into until deleted by the or automatically deleted.

As a hard disk drive (HDD) becomes full, used for data and files will become scattered over different and on the disk surface. A will rearrange the of data and files into sectors wherever possible, thus reducing the scattering of data. It also reduces the number of HDD

Many computers use which automatically launch when a computer has been inactive for a period of time. They form part of the, so that a user is automatically logged out after a certain period of time and the will indicate that the computer is now

Question Number 16**[7]**

a Write down **five** of the functions of a typical operating system.

- 1
- 2
- 3

b The interface between a computer and a user is either a command line interface (CLI) or graphical user interface (GUI).

i Explain the terms CLI and GUI.

CLI:

GUI:

ii In the table below, six statements about interfaces are given. Tick (✓) the appropriate box to indicate whether each statement refers to a CLI or GUI interface. **[6]**

Statement	CLI (✓)	GUI (✓)
The user is in direct communication with the computer		
The user is limited to the icons shown on the screen		
The user needs to learn a number of commands to carry out any operation		
Commands need to be typed in using the correct format and spelling		
There is no need for the user to learn any commands to use the interface		
The interface needs a complex operating system, such as <i>Windows</i> , to operate, which uses considerable amounts of memory		

Question Number 17

a i Explain what is meant by an interrupt.[2]

-
-
-

ii Give **three** examples of what can cause an interrupt.[3]

- 1
- 2
- 3

b i Explain what is meant by a buffer.[2]

.....

ii Explain why buffers are needed.[3]

.....

Memory management, security management and hardware management are three functions of an operating system. The following table shows eight statements. By ticking (✓) the appropriate box, indicate which statements refer to memory, security or hardware management.

Statement	Management type		
	Memory (✓)	Security (✓)	Hardware (✓)
Keeps a track of all memory locations			
Ensures that anti-virus software is regularly updated			
Prevents competing applications using the same memory locations at the same time			
Ensures that the appropriate device driver takes data from memory/file and translates it into a format the device can understand			
Management of devices to control the use of queues and buffers holding data temporarily			
Receives and handles error messages and interrupts from devices connected to the computer			
Manages RAM and allows data to be moved between RAM and devices, such as HDD or SSD			
Maintains access rights for all users of a computer system			

Question Number 19

a Explain the function of a device driver.[1]

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b Explain the use of descriptors in a device driver.[2]

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c Explain the role of a descriptor when a new device is plugged into a USB port of a computer for the first time.[2]

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Question Number 19

11 Use the following list of words and phrases to complete the paragraph below.

Each word or phrase can be used once, more than once or not at all.

assembler	machine code
compilers	programmer
high-level	programs
interpreters	translates
language	translators

Programs written in a low-level are translated into by an

..... before they can be run on a computer. Programs written in a

..... are also translated into before they can be run on a computer.

There are two types of for languages and

.....